

Cognitive Biases in Recommender Systems

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Abstract

Psychology and sociology research has revealed a plethora of cognitive biases that describe systematic deviations from rationality and objectivity in humans' cognitive processes and judgment or collective prejudices of a society that favor one group's values, norms, and traditions over others. Examples such as confirmation bias, conformity bias, or cultural homophily are particularly relevant in the context of recommender systems. These biases are reflected in user-generated data (such as ratings or tags) and, as a consequence, are embedded in user models that drive recommendation algorithms. As a result, recommender systems may reproduce or amplify such biases, leading to undesired outcomes such as stereotypical recommendations, reduced diversity, or unfair exposure. At the same time, cognitive biases shape how users perceive, interact with, and provide feedback to recommender systems, thereby influencing feedback loops and long-term system dynamics. In this multidisciplinary tutorial, we bridge insights from psychology and computer science to examine cognitive biases that are particularly relevant in the context of recommender systems. We introduce these biases grounded in empirical findings and discuss how they manifest across the recommender system pipeline. Participants are engaged through interactive exercises that allow them to experience the effects of cognitive biases firsthand. In a hands-on part, participants are able to (1) design psychology-informed user choice models based on the presented cognitive biases, (2) integrate them in a provided feedback loop simulation setup with established recommendation algorithms, and (3) observe how different biases influence the evolution of user profiles, recommendations, and system performance over time.

CCS Concepts

• **Information systems** → *Recommender systems; Decision support systems*; • **Human-centered computing** → *Collaborative and social computing*; • **Applied computing** → *Psychology*.

Keywords

Psychology, Decision Making, Rationality, Heuristics, Cognition, Bias, Cognitive Effects, Recommender Systems, User Modeling

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1 Motivation and Overview

Human judgment and decision making are strongly influenced by cognitive biases such as conformity bias, confirmation bias, declineism, and halo effect. Extensive research in psychology and sociology has revealed a plethora of such biases that describe *systematic perceptual deviations from rationality and objectivity* in human cognition and perception [2, 3], as well as *collective societal biases* that favor particular values, norms, and groups over others [1, 4].

Recommender systems (RS) play a central role in shaping user decisions in domains such as e-commerce, news consumption, hiring, and entertainment. At their core, RS rely on user-generated data and interactions signals, which inherently reflect human cognitive biases. As a result, such biases can become embedded in user and item representations, learned by ranking models, and ultimately reinforced through system outputs and user feedback loops. This can lead to well-known challenges in RS, including bias amplification, reduced diversity, and unfair or unbalanced exposure of items and providers. Understanding how cognitive biases influence both user behavior and system dynamics is therefore critical for the design, evaluation, and deployment of RS. It enables researchers and practitioners to better interpret observed interaction patterns, identify sources of bias in the recommendation pipeline, and develop methods that mitigate harmful effects.

In this tutorial, we shed light on a mix of well-studied biases (e.g., *primacy/recency effect* and lesser explored ones (e.g., *halo effect*). We revisit their original psychological or sociological definitions and discuss how they manifest in RS across different stages of the pipeline, e.g., data collection, user and item modeling, ranking, and user-system interaction. We also elaborate on the implications for fairness, robustness, and long-term system behavior.

The tutorial addresses an intermediate audience in terms of RS expertise. Knowledge about corresponding fundamentals such as the concepts of implicit and explicit user feedback, ratings, collaborative filtering, or content-based filtering is a requirement. We do not assume any prior knowledge in psychology.

2 Learning Objectives

The purpose of this tutorial is to provide its attendees with an introduction to cognitive biases that are important within the research area of RS. Our goal is to raise awareness of the *psychological aspects*



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pertaining to research, development, and usage of recommendation technology in the RS community. To achieve this, we discuss and intertwine research from the two fields. More precisely, after attending this tutorial, participants will:

- (1) possess a basic knowledge of human decision making theory, corresponding models, notions of rationality, and decision heuristics;
- (2) know the most important cognitive biases and how they affect human decision making;
- (3) understand how and to which extent cognitive biases manifest in different parts of the RS ecosystem;
- (4) possess some ideas about how to leverage negative and mitigate positive effects of cognitive biases in the context RS;
- (5) know the basics of designing and conducting experiments to identify and quantify cognitive biases in a RS scenario.

3 Outline and Structure

The tutorial is organized into four parts: an introduction, a discussion of relevant cognitive biases, a hands-on session, and a reflection on the current state of research and open challenges. In detail, the tutorial covers the following aspects and is structured accordingly:

(1) Introduction

We introduce the tutorial’s motivation, background, presenters, and relevance to the community. We also outline its objectives and overview the covered subject matters.

(2) Cognitive Biases in the Context of RS

We discuss a mixture of well-studied and lesser-studied cognitive biases in the context of RS, pertaining to the user and item representation, the system (training, model, inference), and the user-system interactions. Concrete topics include:

- (a) *Rationality, utility theory, decision heuristics*
- (b) *The role of cognitive biases in human decision making*
- (c) *Primacy/Recency effects and position bias*: The position at which an item appears in a list influences its received attention. Humans tend to recall first and last items from a sequence better than items appearing in the middle.
- (d) *Bandwagon effect and popularity bias*: Humans tend to adopt a certain behavior or belief because many others behave or believe in that way.
- (e) *Feature-positive effect*: Humans are better at realizing the presence of a stimulus rather than its absence.
- (f) *IKEA effect*: The more effort a person invested into something, the more they value it.
- (g) *Cultural homophily*: Humans tend to form connections with others who have similar traits more often than with others who have different traits.
- (h) *Conformity bias*: Humans commonly align their beliefs, behaviors, and actions with those of a group.
- (i) *Declinism*: People often perceive that the world or society is declining, i.e., things were better in the past and have become worse over time.
- (j) *Halo and horn effects*: A person’s perception of a single attribute of another person or thing influences their opinion on unrelated attributes, in a positive or negative way.
- (k) *Strategies to mitigate and leverage cognitive biases in RS*

(3) Hands-on Experiments

- (a) Introduction of the feedback loop simulation setup, based on the framework the tutorial instructors presented in [8].
- (b) Formalizing and implementing cognitive effects as user choice models.
- (c) Running recommendation feedback loop simulations and analyzing the progression of the system states (user profile, recommendations, and item exposure deviations).

(4) Reflection and Open Challenges

We wrap up with a discussion of opportunities and challenges, including:

- (a) Challenges in designing and conducting experiments to transfer findings from behavioral psychology to the tasks of user- and item modeling and recommendation.
- (b) Mitigating negative and leveraging positive effects of cognitive biases in a recommendation context.
- (c) Raising a broader awareness of potential ways cognitive biases can be exploited, e.g., to manipulate people.
- (d) Improving the communication between the relevant communities, including computer science, artificial intelligence, psychology, and sociology, to foster interdisciplinarity.

4 Supporting Materials

The presentation slides and hands-on resources are available at https://cpjku.github.io/cobis_rs_recsys2026. For the interactive part, attendees are encouraged to bring their own laptops. The hands-on experiments run with standard Python environments (e.g., via Jupyter Notebook or Google Colab). The provided notebooks and datasets are lightweight and self-contained, enabling participants to replicate and extend the experiments during and after the tutorial.

5 Biographies and Relevant Publications

Markus Schedl is a full professor at the Johannes Kepler University (JKU) Linz, affiliated with the Institute of Computational Perception (ICP), leading the Multimedia Mining and Search Group and the Human-centered AI Group at the Linz Institute of Technology, AI Lab. His research interests include information retrieval, recommender systems, and trustworthy AI, with a particular focus on detecting and mitigating bias in retrieval and recommendation algorithms [10, 14, 15, 17] and on psychological aspects of the corresponding processes [12, 13, 16, 18].

Oleg Lesota is a final-year PhD student at the ICP of the JKU. His research revolves around the user side of recommender systems, including calibrating recommendations to individual user preferences [5–7, 10], cultural impacts of recommender systems [8, 11], and cognitive effects in recommendation and HCI [9, 18].

Antonela Tommasel is an assistant professor at JKU and at the UNCPBA, as well as a researcher at the CONICET. Her research lies at the intersection of artificial intelligence and social computing, with a focus on recommender systems. Her recent work examines fairness and biases in recommender systems, including LLM-based systems [22] in group recommendation settings [20, 21] and investigating all stages of the recommendation pipeline [20]. She has also examined how positional effects can influence LLM outputs [19], with implications for the reliability and evaluation of LLM-driven recommendations.

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